1. Please use the following system of linear equations to answer the next 2 questions (1.1 and 1.2.).

$$\begin{aligned} 3x_1 + 5x_2 - 3x_3 &= 10, \\ 4x_2 + 4x_3 &= -5, \\ 2x_1 - 9x_2 + 6x_3 &= 13. \end{aligned}$$

1.1. Use LU Decomposition Method to get the matrices, L and U. (20 pts)

1.2. Write the Python code to solve the system shown above. You can use any one of the methods studied in the class. (20 pts)

2. Please use power method to find the approximated largest eigenvalue of the matrix $\begin{bmatrix} 0 & 3 \\ 3 & 2 \end{bmatrix}$. You can use the initial vector [1,1]. (show 3 iterations) (20 pts)

3. Given the following information, write a Python code for computing parameters of the estimation function by using the Least Squares Regression method. (20 pts)

```
x = np.linspace(0, 1, 101)
y = 1 + x * np.random.random(len(x))
```

estimation function $\hat{y}$: $\hat{y} = \alpha_1 x + \alpha_2$

4. Find the linear interpolation at $x = 3$ based on the data $x = [2, 5, 6]$ and $y = [2, 3, 1]$. (20 pts)

5. Assume that we are using Cubic Spline Interpolation. Given the following data points $x = [2, 5, 6]$ and $y = [2, 3, 1]$, please find 8 cubic functions to set up the system of linear equations. (20 pts)

(hint: if $f(x) = ax^3 + bx^2 + cx + d$, $f(x)' = 3ax^2 + 2bx + c$ and $f(x)'' = 6ax + 2b$)

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6. Given the following data points $x = [2, 5, 6]$ and $y = [2, 3, 1]$, please build 3 polynomials for Lagrange Polynomial Interpolation method. (20 pts)

7. Write a Python code to find an approximated root of the function $f(x) = x^2 + 3x + 1$ using Newton-Raphson method. Please explain each line what it does/means. Use $x_0 = 3$ as a starting point and `1e5` as a tolerance. (hint: $f(x)' = 2x + 3$ and `newton_raphson` can be written as a recursive function.) (40 pts)